INCH-POUND

MIL-C-43348A(GL)
6 September 1990
SUPERSEDING
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14 July 1965

MILITARY SPECIFICATION

CHALK, MARKING, REFLECTIVE

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a reflex-reflective material in chalk form that can be applied to clothing and can later be removed (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7510

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SPECIFICATIONS

FEDERAL

PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing
PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-636 - Box, Shipping, Fiberboard
PPP-B-676 - Boxes, Setup

MILITARY

MIL-C-10296 - Cloth, Sateen, Cotton
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5094.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.1.1 Materials. The reflective chalk shall be made from spherical glass elements which are specially treated to reflect light and are bonded with a suitable organic binding material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.1.1.1 Wrapper. The wrapper for the chalk shall be suitable for holding the chalk firmly and for protecting the enclosed chalk and shall be easily removed or unwound in small increments from the chalk.

3.2 Dimensions. The reflective chalk shall be circular in cross section with a diameter of 7/16 inch, minimum, and a length of 1-1/2 inches ± 1/8 inch.

3.3 Weight. The weight of a piece of wrapped chalk shall be 10.8 grams minimum, when tested as specified in 4.4.4.

3.4 Reflective intensity. The reflective intensity of the chalk shall be as specified in table I when tested as specified in 4.4.4.

TABLE I. Reflective intensity at an angle of divergence of 0.2°

Angle of incidence	-4°	0°	10°	20°	30°	40°	50°
	5.0	5.0	5.0	4.5	4.0	3.5	3.0

3.5 Color. The chalk shall be gray in color.

3.6 Workmanship. The chalk shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of chalk. The sample units shall be one piece of chalk and not more than one sample unit shall be sampled from each package. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

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TABLE II. End item visual defects

Examine	Defect
Color	Not gray color Not uniform
Appearance	Shape not cylindrical Any break, crack, or chip (except small chips on periphery of either end) Wrapper missing, torn, not easily unwound in small increments Wrapper not continuous or too loosely affixed, causing chalk to fall or slide from wrapper
Application	Does not leave mark when applied to dark clothing or cloth

4.4.3 End item dimensional examination. The end items shall be examined for the dimensions specified in 3.2. The lot size shall be expressed in units of chalk. The sample unit shall be one piece of chalk and not more than one sample unit shall be sampled from each package. The inspection level shall be I and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.4 End item testing. The end items shall be tested for the characteristics listed in table III. The lot size shall be expressed in units of packages of chalk (20 pieces) each. The sample unit shall be one package of chalk randomly selected throughout the lot. The sample size shall be as shown below. All test characteristics are applicable to the sample unit. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified in table III.

Lot size (packages of 20 pieces of chalk)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE III. End item tests

Characteristic	Requirement	Test method	Number of determinations per sample unit	Results reported as numerically to the nearest
Weight	3.3	4.5.1	average of 20	0.1 gm
Reflective intensity	3.4	4.5.2	average of 20	0.1

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number of intermediate containers is more or less than specified Number of chalk per intermediate container is more or less than specified

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Weight. Weigh 20 randomly selected pieces of reflective chalk on a laboratory balance accurate to 0.1 gram. The average weight shall conform to the requirement in 3.3.

4.5.2 Reflective intensity.

4.5.2.1 Panel preparation. Tape a 7 by 7 inch square piece of cloth conforming to MIL-C-10296, with the face (filling effect side) exposed to an 8 by 8 inch square clean, smooth metal panel to that the cloth is held taut around all edges and no metal is exposed. Final exposed area of the cloth, after taping, should be 6 by 6 inches square. Flatten the end of the chalk specimen on No. 400 grit sandpaper. Hold the flattened end of the chalk against the cloth and make a series of individual adjoining marks in one direction from the top of the panel to the bottom. When the 6 by 6 inch square area has been filled in with chalk marks, make one mark each across the top and bottom edges at right angles to the previously applied marks. Carefully remove chalk from tape edges. Handle the panels carefully, keep in a horizontal position and do not jar or stack.

4.5.2.2 Apparatus. A projector, having a maximum lens diameter of 1 inch and capable of projecting a uniform beam of light with an angular spread not more than 1/6 degree shall be used to illuminate a vertical panel. The illumination source shall be a Tungsten lamp of 2850°K minimum color temperature.1/ The light reflected from the panel shall be measured with a photoelectric receiver whose response has been corrected for the color sensitivity of the eye. A selenium barrier layer photocell corrected to the sensitivity of the eye with a Kodak Wratten Filter No. 102 shall be considered proper color correction for the purpose of this test. The maximum diameter of the active area of the receiver shall be 1 inch. The distance from the center of the vertical panel to both the front lens of the projector and the receiver shall be 50 feet ± 2 inches. The arrangement shall be as shown in figure 1.

1/ A 500 or 1000 watt movie projector has been used as the light source.

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4.5.2.3 Test procedure. Measure the distance from the projector to the panel, the area of the panel (0.25 sq. ft.) and the illumination incident on the panel. Measure the illumination incident on the receiver due to reflection from the vertical panel at the seven angles of incidence[1/] for the 0.2° angle of divergence.[2/] The illumination incident on the panel and the receiver shall be measured in the same units and may be measured by the same receiver. Compute the reflective intensity R, from the following equation:

$$R = \frac{L_r D^2}{L_p A}$$

Where R = Reflective intensity.

L_r = Illumination incident upon the receiver.

L_p = Illumination incident upon a plane perpendicular to the incidence ray at the panel position, measured in the same units as L_r.

D = Distance in feet from the panel to the projector.

A = Area in square feet of the panel = 0.25 sq. ft.

The reflective intensity shall be determined on 20 randomly selected pieces of chalk. The average value shall be not less than the minimum requirements specified in 3.4, table I.

1/ The incidence angle has as its sides the lines normal to the center of the panel and the line joining the center of the panel to the center of the projection lens. If the line normal to the panel is on the same side of the projector as the receiver, the incidence angle is positive. If the line normal to the panel is on the opposite side of the projector from the receiver, the incidence angle is negative.

2/ The divergence angle has at its sides the line joining the center of the panel with the center of the projection lens and the line joining the center of the panel with the center of the receiver.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation. Twenty pieces of chalk shall be unit packed in a snug-fitting folding paperboard box conforming to variety 1, style II, type D, class e or style III, type G, class i or class j of PPP-B-566; or setup paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676. Closure of telescope type boxes shall be secured with gummed paper tape conforming to type III, grade B of PPP-T-45.

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5.1.2 Commercial preservation. The chalk shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Two hundred boxes of chalk, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Inside dimensions of the box shall be approximately 13 inches in length, 10-1/2 inches in width, and 13 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type III, class F loads.

5.2.2 Level B packing. Two hundred boxes of chalk, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Inside dimensions of the box shall be approximately 13 inches in length, 10-1/2 inches in width, and 13 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.3 Commercial packing. The chalk, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), chalk, packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 16 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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5.4.1 Precautionary marking. The following precautionary marking shall appear on one end and side panel of each shipping container in capital letters not less than 3/4 inches in height, as follows:

STORAGE CONDITIONS SHALL NOT
EXCEED 95% RELATIVE HUMIDITY

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The reflective chalk is intended for temporary markings on clothing of uniformed personnel and other objects that require temporary reflectorization where use of other reflective media are not practical or convenient. When properly applied, the chalk provides a brightly visible silver line or area when viewed in a light beam in the darkness. Best applications are made on cloth, paper, wood, rubber, and other textured or porous materials; however, satisfactory applications have been made on metals, rocks, and most non-enamel painted surfaces. Application to very smooth, solid surfaces will result in larger coverage areas; however, brightness values will be less. The markings can be removed from clothing by normal washing or dry cleaning methods and from other surfaces by washing or wiping with a damp cloth.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Size and shape (see 3.2).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.4 Subject term (key word) listing.

Marker, cloth
Marker, nonpermanent

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Review activity:

Army - MD

Civil agency coordinating activity:

GSA - FSS

Preparing activity:

Army - GL

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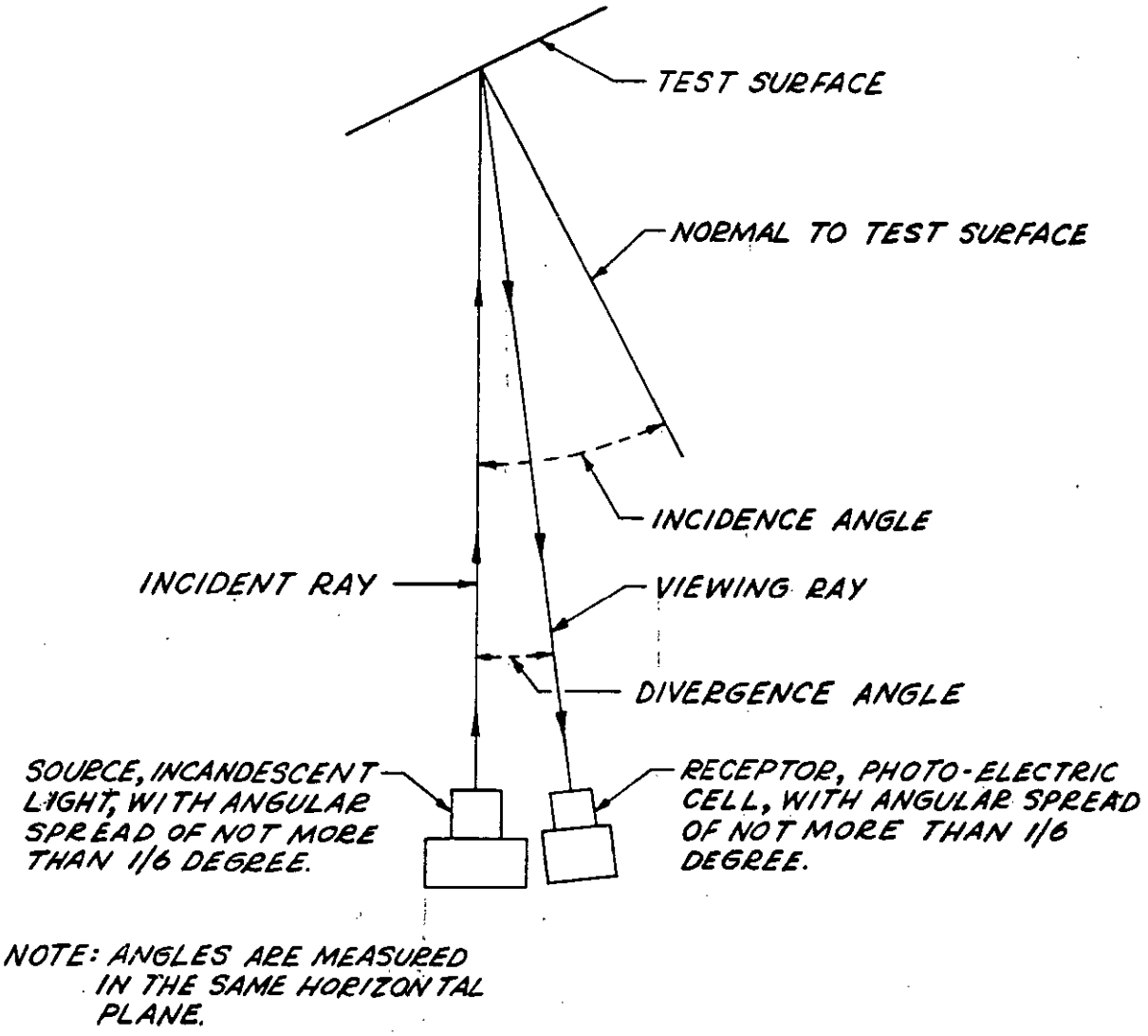


FIGURE 1. Incidence and divergence angles as measured in reflection factor test.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
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3. DOCUMENT TITLE

Chalk, Marking, Reflective

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Natick RD&E Center	508-651-5221	256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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